

SPECTRAL LIKELIHOOD, PROJECTIVE CRITICAL SCHEMES, AND RAMIFICATION FOR QUADRATIC PENCILS

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ABSTRACT. This separately reviewable applications manuscript retains the reciprocal likelihood and projective critical-scheme theory of quadratic pencils. For supplied split semisimple spectral data with simple disjoint projective poles, the critical divisor is finite locally free through critical collisions; its ramification ideal equals its Hessian ideal. For a supplied real definite realization, separated real data give a totally real critical cover. The score discriminant is not inverted in the finite-flat theorem. Real structure and data are additional inputs, not conclusions of complex unmarked reconstruction. References to the finite-failure inverse and spectral readout refer to the separate principal article; neither this application nor its real-root proof is an input to that article.

A1. A TOTALLY REAL RECIPROCAL LIKELIHOOD FIBRE

The spectral data also lead to a question independent of reconstruction: can every critical point of reciprocal likelihood on a definite real pencil be made real by a choice of real data? The assertion was formulated as Conjecture 4.5 in [1]. We give an explicit construction valid for every number of distinct eigenvalues and every multiplicity pattern. Its proof counts the critical points directly; it does not assume the complex likelihood-degree formula.

A real symmetric pencil containing a positive definite matrix can be written, after a real congruence and a choice of generators, as $\langle I, \text{diag}(a_1, \dots, a_n) \rangle$. Indeed, make the definite matrix the identity and orthogonally diagonalize the other symmetric matrix. Group its distinct eigenvalues as $\alpha_1 < \dots < \alpha_r$, with multiplicities $m_1, \dots, m_r > 0$ and $n = \sum_i m_i$. A two-dimensional pencil has $r \geq 2$. If σ_i is the sum of the data entries in the i -th eigenspace, the reciprocal score equations are the critical equations of

$$\tilde{\ell}(x, y) = - \sum_{i=1}^r m_i \log(\alpha_i x + y) - \sum_{i=1}^r \frac{\sigma_i}{\alpha_i x + y} \quad (\text{A1.1})$$

on the complement of $\prod_i (\alpha_i x + y) = 0$. The derivatives are rational and independent of the local branches of the logarithms. On the real complement one can equivalently use $\log |\alpha_i x + y|$.

Theorem A1.1 (Explicit totally real likelihood data). *Choose any real numbers*

$$\alpha_r < \beta_1 < \dots < \beta_{r-1}, \quad D(t) = \prod_{i=1}^r (t - \alpha_i), \quad Q(t) = \prod_{j=1}^{r-1} (t - \beta_j),$$

and prescribe

$$\sigma_i = \frac{Q(\alpha_i)}{D'(\alpha_i)}. \quad (\text{A1.2})$$

Then (A1.1) has exactly $2r - 3$ critical points on its complex domain. Every one is real and nondegenerate. There is one critical point with $t = -y/x$ in each interval (α_i, α_{i+1}) , and one in each interval (β_j, β_{j+1}) ; there are no others. The same conclusion holds on a nonempty Euclidean

open subset of the real data space. Thus the assertion of Conjecture 4.5 in [1] holds for all definite real pencils.

The data here are unrestricted real data, exactly as in that assertion. Neither positivity of the data matrix nor the assertion that all the critical points lie in the positive definite cone is part of the theorem. For example, the residues in (A1.2) alternate in sign.

Lemma A1.2 (Two separated strings of roots). *With the notation of the theorem, put*

$$H(t) = \sum_{i=1}^r (n - m_i) \frac{D(t)}{t - \alpha_i}, \quad F(t) = nD(t)Q'(t) - H(t)Q(t). \quad (\text{A1.3})$$

The polynomial F has degree exactly $2r - 3$. It has a simple root in each of the intervals in the theorem, has no other root over $\mathbb{C}$, and is relatively prime to DQ .

Proof. Since $\sum_i (n - m_i) = n(r - 1)$, the terms of degree $2r - 2$ in F cancel. At a zero of D ,

$$F(\alpha_i) = -(n - m_i)D'(\alpha_i)Q(\alpha_i).$$

The factors $n - m_i$ are positive, all the values $Q(\alpha_i)$ have the same sign, and consecutive values $D'(\alpha_i)$ have opposite signs. Thus F changes sign in each of the $r - 1$ intervals between the α_i . At a zero of Q ,

$$F(\beta_j) = nD(\beta_j)Q'(\beta_j).$$

Here $D(\beta_j) > 0$, and consecutive values of Q' have opposite signs. There are therefore a further $r - 2$ roots in the intervals between the β_j . These intervals are pairwise disjoint and are disjoint from all the former intervals. The upper bound $2r - 3$ on the degree forces exactly one root in each interval, all simple, and excludes any remaining real or complex root. The displayed endpoint values are nonzero, proving coprimality. This argument includes $r = 2$, when the second list of intervals is empty. $\square$

Proof of Theorem A1.1. Partial fractions give

$$\frac{Q(t)}{D(t)} = \sum_i \frac{\sigma_i}{t - \alpha_i}, \quad S(t) := \sum_i \frac{\sigma_i}{\alpha_i - t} = -\frac{Q(t)}{D(t)}. \quad (\text{A1.4})$$

On the chart $x \neq 0$, put $y = -xt$. Locally, up to a constant, the function becomes

$$-n \log x - \sum_i m_i \log(\alpha_i - t) - \frac{S(t)}{x}.$$

The first critical equation gives $x = S(t)/n$. If $S(t) = 0$ it cannot be satisfied on this chart. When $S(t) \neq 0$, the remaining critical equation is

$$\left. \frac{\partial \ell}{\partial t} \right|_{x=S(t)/n} = -\sum_i \frac{m_i}{t - \alpha_i} - n \frac{S'(t)}{S(t)} = -\frac{F(t)}{D(t)Q(t)} = 0. \quad (\text{A1.5})$$

The lemma now gives exactly $2r - 3$ solutions in this chart, with the real coordinates

$$x = -\frac{Q(t)}{nD(t)}, \quad y = -xt. \quad (\text{A1.6})$$

At each of these points $x \neq 0$ and $\alpha_i x + y \neq 0$, so none is an extraneous solution introduced by clearing denominators.

We must also exclude the omitted chart $x = 0$. Comparing the coefficients of t^{-1} and t^{-2} in (A1.4) gives

$$\sum_i \sigma_i = 1, \quad \sum_i \alpha_i \sigma_i = \sum_i \alpha_i - \sum_j \beta_j.$$

At $x = 0$, the y -equation would imply $y = 1/n$. The x -equation would then require

$$\sum_i (n - m_i) \alpha_i = n \sum_j \beta_j. \quad (\text{A1.7})$$

But the left side is strictly smaller than $n(r - 1)\alpha_r$, whereas the right side is strictly larger. Thus there is no critical point on $x = 0$. This proves the assertion about the entire complex domain, not just the chosen coordinate chart.

For nondegeneracy, at a critical point in (x, t) -coordinates one has $\tilde{\ell}_{xx} = -n/x^2 \neq 0$. The derivative of the eliminated equation (A1.5) is $-F'(t)/(D(t)Q(t)) \neq 0$. It is the Schur complement of $\tilde{\ell}_{xx}$ in the Hessian. Hence the Hessian is nonsingular. The coordinate change $(x, t) \mapsto (x, -xt)$ has nonzero Jacobian at these points, so nondegeneracy also holds in the original coordinates.

Finally allow Q to have any positive leading coefficient and $r - 1$ distinct real roots to the right of α_r . These polynomials form a nonempty Euclidean open set in their r -dimensional coefficient space. The linear map $Q \mapsto (Q(\alpha_i)/D'(\alpha_i))_{i=1}^r$ is an isomorphism, by polynomial interpolation. Scaling Q does not change its critical t -values and scales (x, y) by the same nonzero factor. Thus its image is the required open set of grouped data. The map from the original n data entries to their r group sums is surjective, so its inverse image is open and nonempty as well. $\square$

Example A1.3. For $(\alpha_1, \alpha_2, \alpha_3) = (1, 2, 3)$, $m_1 = m_2 = m_3 = 1$, and $(\beta_1, \beta_2) = (4, 5)$, the data are $(6, -6, 1)$ and

$$F(t) = 15t^3 - 130t^2 + 345t - 278.$$

Its three roots lie respectively in $(1, 2)$, $(2, 3)$, and $(4, 5)$. The theorem proves the same assertion for arbitrary r , not just for examples reached by root finding.

Remark A1.4 (Relation to the intrinsic invariant). The finite failure neighbourhood recovers the complex projective pencil and its spectral sheaf. To apply the real theorem one specifies a real structure and a definite member; these are not asserted to be recoverable from an unmarked complex scheme. After that specification, the construction uses only the ordered spectral points, their multiplicities, and the freely chosen β_j . The result is an independently stated consequence in reciprocal likelihood geometry, rather than a new name for Weierstrass–Segre classification. It proves the cited conjectural assertion without claiming that every real critical point is a statistically admissible maximum.

A2. THE RECONSTRUCTED CRITICAL CORRESPONDENCE

This section identifies the algebraic object joining the finite inverse theorem to the real likelihood result. The reconstruction step is a functorial consequence of recovering the pencil. The additional content is a coordinate-free data construction and a relative critical algebra whose real points split into globally labelled sections.

A2.1. The universal score scheme. For matrix formulas we regard quadrics as symmetric bilinear forms on a vector space; this amounts to dualizing the vector-space convention used earlier. Write $\mathcal{W} = \text{Sym}^2 V^*$. For a regular pencil $R \subset \mathcal{W}$, let $R^\times = \{K \in R : \det K \neq 0\}$. A data matrix S belongs to the same space $\mathcal{W}$. Define a section of the trivial bundle with fibre R^* on $R^\times \times \mathcal{W}$ by

$$\eta_{K,S}(H) = -\text{tr}(K^{-1}H) + \text{tr}(SK^{-1}HK^{-1}), \quad H \in R. \quad (\text{A2.1})$$

Its zero scheme $\mathfrak{C}_R$ is the reciprocal critical correspondence, with its projection to the data space $\mathcal{W}$. Over the reals these are the critical equations of $-\log |\det K| - \text{tr}(SK^{-1})$; the scheme itself uses no choice of logarithm and is defined over the complex field.

Theorem A2.1 (Reconstruction and base change of the critical correspondence). *Let $n \geq 3$ and $d = n^2 + 2n - 4$. For every regular pencil R , the abstract scheme $Z_R^{[d]}$ determines the isomorphism class of the correspondence*

$$\mathfrak{C}_R \longrightarrow \mathcal{W}$$

up to simultaneous congruence on the pencil and data space. On the regular pencil parameter scheme the score sections define a universal zero scheme. Its pullback along any complex base change is the corresponding relative critical scheme. The classifying map of a family in the framed first-relation locus therefore recovers this relative correspondence, including over nonreduced bases. No chosen data matrix or real form is part of the unmarked conclusion.

Proof. If g is invertible, put $K' = g^\dagger K g$, $H' = g^\dagger H g$, and $S' = g^\dagger S g$. Cyclic invariance of trace gives

$$\eta_{K', S'}(H') = \eta_{K, S}(H).$$

Thus simultaneous congruence identifies the score sections and their zero ideals, not just their closed points. By Theorem 10.1, an isomorphism of the finite neighbourhoods recovers the pencil up to projective congruence. A linear lift of that projective transformation supplies the displayed isomorphism of correspondences. Different lifts differ by a common scalar change of coordinates on the pencil and data space, so they leave its isomorphism class unchanged.

On $G^{\text{reg}} \subset \text{Gr}(2, \mathcal{W})$, let $\mathcal{R}$ be the tautological pencil bundle. On the determinant open in $\text{Tot } \mathcal{R}$, inversion of the universal form is a regular operation. Formula (A2.1) is a section of the pullback of $\mathcal{R}^*$ after taking the product with $\mathcal{W}$. In a frame it has two entries. Their generated ideal is independent of the frame and its quotient commutes with arbitrary base change, as does localization at the determinant. This proves the relative assertion directly, without an assertion that these score schemes are flat everywhere. Finally the closed immersion of Theorem 10.3 identifies a framed first-relation family with its classifying morphism to G . Pulling back the universal score construction proves the last assertion. It does not identify unmarked moduli stacks. $\square$

A2.2. Spectral projectors and a congruence-covariant data section. Suppose now that A is real symmetric positive definite, B is real symmetric, and $R = \langle A, B \rangle$ is a pencil. Set $C = A^{-1}B$. Its distinct eigenvalues and multiplicities are

$$\alpha_1 < \cdots < \alpha_r, \quad m_1, \dots, m_r, \quad \sum_i m_i = n,$$

where $r \geq 2$. Its spectral idempotents are

$$P_i = \prod_{j \neq i} \frac{C - \alpha_j I}{\alpha_i - \alpha_j}, \quad P_i P_j = \delta_{ij} P_i, \quad \sum_i P_i = I. \quad (\text{A2.2})$$

They are selfadjoint for A . Define the grouped data map

$$\tau_C : \mathcal{W}_{\mathbb{R}} \longrightarrow \mathbb{R}^r, \quad \tau_C(S)_i = \text{tr}(P_i A^{-1} S). \quad (\text{A2.3})$$

For $K = xB + yA$, matrix functional calculus and cyclic trace give

$$-\log |\det K| - \text{tr}(SK^{-1}) = -\log \det A - \sum_i m_i \log |\alpha_i x + y| - \sum_i \frac{\tau_C(S)_i}{\alpha_i x + y}. \quad (\text{A2.4})$$

Thus the entire score scheme, not only its critical count, depends on S through the surjective linear map τ_C .

Choose $c > 0$ and $\alpha_r < \beta_1 < \cdots < \beta_{r-1}$, and set

$$\begin{aligned} D(t) &= \prod_i (t - \alpha_i), & Q(t) &= c \prod_j (t - \beta_j), \\ \sigma_i &= \frac{Q(\alpha_i)}{D'(\alpha_i)}, & S_0 &= A \sum_i \frac{\sigma_i}{m_i} P_i. \end{aligned} \tag{A2.5}$$

The form S_0 is symmetric and $\tau_C(S_0) = \sigma$, since $\text{tr } P_i = m_i$. More generally the assignment $\sigma \mapsto A \sum_i (\sigma_i/m_i) P_i$ is a right inverse of τ_C ; all data with these grouped values are $S_0 + \ker \tau_C$. Under simultaneous congruence, $C \mapsto g^{-1} C g$, $P_i \mapsto g^{-1} P_i g$, and $S_0 \mapsto g^\dagger S_0 g$. No eigenbasis has been chosen.

Theorem A2.2 (A totally real finite etale critical cover). *Fix the multiplicities $m_1, \dots, m_r$. On the locus of supplied real definite pairs (A, B) with these ordered multiplicities, there is a nonempty fibrewise open semialgebraic set of data defined by the following condition: the polynomial*

$$Q_S(t) = \sum_i \tau_C(S)_i \frac{D(t)}{t - \alpha_i} \tag{A2.6}$$

has positive leading coefficient and $r - 1$ distinct real roots, all greater than α_r . On this set the reciprocal critical scheme has exactly $2r - 3$ points, all real and nondegenerate. They form globally labelled real analytic semialgebraic sections: $r - 1$ have their spectral coordinate in (α_i, α_{i+1}) , and $r - 2$ in the gaps between successive roots of Q_S .

In algebraic spectral coordinates the same family is the restriction of a finite etale algebra of rank $2r - 3$ over a specified Zariski open base. Its construction commutes with arbitrary base change on that base. The formula in (A2.5) provides a congruence-covariant section of the data map, and its entire affine fibre has the same critical scheme.

Proof. Write $Q = Q_S$, and put

$$H(t) = \sum_i (n - m_i) \frac{D(t)}{t - \alpha_i}, \quad F(t) = nD(t)Q'(t) - H(t)Q(t). \tag{A2.7}$$

Both apparent leading terms have degree $2r - 2$ and cancel. Writing $Q = c \prod_j (t - \beta_j)$, the coefficient of t^{2r-3} is

$$a = c \left(n \sum_j \beta_j - \sum_i (n - m_i) \alpha_i \right) > 0. \tag{A2.8}$$

The strict inequality follows from $\beta_j > \alpha_r$, $n - m_i > 0$, and $\sum_i (n - m_i) = n(r - 1)$. Hence F has degree exactly $2r - 3$.

At α_i and β_j one has

$$F(\alpha_i) = -(n - m_i) D'(\alpha_i) Q(\alpha_i), \quad F(\beta_j) = n D(\beta_j) Q'(\beta_j).$$

The signs alternate along each string. The first string gives a root in every (α_i, α_{i+1}) , and the second gives one in every (β_j, β_{j+1}) . These $2r - 3$ disjoint intervals exhaust the degree. All roots are simple and F is coprime to DQ . This also treats $r = 2$, when the second string of intervals is empty.

We next prove the scheme statement and exclude the missing coordinate chart uniformly. Take the polynomial ring over $\mathbb{Q}$ in $\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{r-1}, c$, and localize at c , the products of distinct spectral and β -differences, a , $\text{disc}_t F$, and $\text{Res}_t(F, DQ)$. Call this ring $\mathcal{B}$. Every point of the separated real chamber lies in $\text{Spec } \mathcal{B}(\mathbb{R})$, by the preceding sign argument. For a linear F , its discriminant is understood as 1.

The partial fraction identity $Q/D = \sum_i \sigma_i/(t - \alpha_i)$ shows that

$$\sum_i \sigma_i = c, \quad \sum_i \alpha_i \sigma_i = c \left(\sum_i \alpha_i - \sum_j \beta_j \right).$$

On the chart $x = 0$ of the invertible-pencil locus the score in y forces $y = c/n$. Substitution in the score in x then forces $a = 0$. Since a is inverted, the quotient of the critical coordinate ring by (x) is zero. Thus x is a unit on the whole critical scheme, including nilpotent test rings; excluding this chart is not merely a pointwise argument.

Set $t = -y/x$ and $U(t) = \sum_i \sigma_i/(\alpha_i - t) = -Q(t)/D(t)$. In these coordinates the grouped function is $-n \log x - \sum_i m_i \log(\alpha_i - t) - U(t)/x$, with the logarithms used only to write its rational differential. The score in x gives $x = U/n$; the remaining score becomes $-F/(DQ)$. Consequently the critical coordinate algebra is

$$\mathcal{B}[t]/(F/a), \quad x = -\frac{Q(t)}{nD(t)}, \quad y = \frac{tQ(t)}{nD(t)}. \quad (\text{A2.9})$$

The resultant makes D and Q units in this algebra, so these formulas and the coordinate change are genuine inverse homomorphisms. The polynomial F/a is monic of degree $2r - 3$; its discriminant is a unit. The algebra is therefore finite free of that rank, and its derivative is a unit modulo the polynomial, which proves that it is étale. These algebraic identities persist under every base change of $\mathcal{B}$. The statement is local in spectral coordinates; no global flatness of the unrestricted critical correspondence is inferred.

At a critical point the second derivative in the eliminated x direction is $-n/x^2$. The complementary Schur factor is $-F'(t)/(D(t)Q(t))$. Both are nonzero, so the Hessian is nondegenerate. The change from (x, t) to (x, y) has Jacobian determinant $-x \neq 0$, preserving this conclusion. Each specified real interval contains a unique root. The implicit function theorem therefore supplies a real analytic root function locally, and uniqueness glues these functions globally with the stated labels. Their graphs are semialgebraic, since they are specified by a polynomial equation and the interval inequalities. The ordered roots β_j of Q_S have the same local analytic and global semialgebraic property.

For fixed α , the map from the r coefficients of Q to σ is a linear isomorphism, with inverse (A2.6). Polynomials with the specified simple separated roots and positive leading coefficient form a nonempty open semialgebraic coefficient set. Surjectivity and the explicit right inverse of τ_C make its preimage in the full data space nonempty and open. Spectral projectors give the same formulas along any real analytic family in the fixed multiplicity locus. Their congruence covariance was verified above; the kernel directions of τ_C do not occur in the score by (A2.4). This completes all the assertions. $\square$

Corollary A2.3 (The finite-failure-to-critical construction). *An order- d finite failure invariant of a regular complex pencil, together with a supplied real definite realization and the spectral choices in (A2.5), determines the above finite étale critical family up to simultaneous congruence. The all-real critical count and labelled real sections are independent of an eigenbasis. The real choices are additional inputs, not outputs of complex unmarked reconstruction.*

Proof. Apply Theorem A2.1, use the spectral sheaf to read the eigenvalues and their multiplicities in the supplied definite realization, and apply Theorem A2.2. Equations (A2.2)–(A2.5) verify invariance under all changes of eigenbasis and simultaneous congruence. The proof of total reality is an additional spectral argument, not a premise of the inverse theorem. $\square$

A3. THE FINITE FLAT CRITICAL DIVISOR AND ITS RAMIFICATION

The critical cover of Section A2 was written on an affine spectral chart and away from its discriminant. We now retain the full projective critical divisor. It remains finite and flat when critical directions cross the omitted chart or collide. Its nonreduced fibres and its ramification are recovered from the same finite failure invariant. Thus the spectral construction carries more than the generic critical count or its real realization.

A3.1. The projective score differential. Fix positive integers $m_1, \dots, m_r$, with $r \geq 2$ and $n = \sum_i m_i$. On the split semisimple locus write $K = xB + yA$, where A is nonsingular and $A^{-1}B$ has distinct spectral values α_i with these multiplicities. Over a nonreduced base, this means that mutually orthogonal idempotent projectors P_i of ranks m_i are supplied, with $\sum_i P_i = 1$ and $A^{-1}B = \sum_i \alpha_i P_i$. It is stronger than semisimplicity of the geometric fibres alone. The base field is $\mathbb{C}$; all the universal formulas below are already defined over $\mathbb{Q}$. Put

$$\begin{aligned} L_i &= \alpha_i x + y, & D_h &= \prod_i L_i, & D_i &= \prod_{j \neq i} L_j, \\ Q_h &= \sum_i \sigma_i D_i, & U(K) &= \text{tr}(SK^{-1}) = Q_h/D_h. \end{aligned} \tag{A3.1}$$

Here $\sigma_i = \text{tr}(P_i A^{-1} S)$, with the projectors from (A2.2). The polynomial Q_h is a binary form of degree $r - 1$; its affine leading coefficient is allowed to vanish.

Let $\mathcal{U}$ be the open subscheme of $\text{Spec } \mathbb{Q}[\alpha_1, \dots, \alpha_r, \sigma_1, \dots, \sigma_r]$ on which the α_i are pairwise distinct, Q_h has $r - 1$ distinct projective zeros, and these zeros are disjoint from those of D_h . Equivalently, invert the spectral differences, the binary discriminant of Q_h , and $\text{Res}(D_h, Q_h)$. The discriminant of a binary linear form is understood as 1; the resultant condition then also excludes the zero form. No discriminant of a score polynomial is inverted. In particular, this base allows critical-point collisions. All subsequent base changes are from $\mathcal{U}$.

Consider the following rational one-form on the projective line:

$$\begin{aligned} \omega &= \sum_i (n - m_i) d \log L_i - n d \log Q_h = \frac{\eta}{D_h Q_h}, \\ \eta &= Q_h \sum_i (n - m_i) D_i dL_i - n D_h dQ_h. \end{aligned} \tag{A3.2}$$

Differentials are relative to the base. The equality $\sum_i (n - m_i) = n(r - 1)$ makes this expression independent of rescaling a representative (x, y) . If $\eta = \eta_x dx + \eta_y dy$, Euler's identity gives $x\eta_x + y\eta_y = 0$. The syzygy of (x, y) therefore gives a unique binary form G of degree $2r - 3$ such that

$$\eta = G(x, y)(x dy - y dx). \tag{A3.3}$$

This argument is valid over the universal coefficient ring: reducing $x\eta_x + y\eta_y = 0$ modulo x first shows that x divides η_y , and then determines both coefficients. It remains valid after every base change.

On the chart $(x, y) = (1, -t)$, let $D(t) = \prod_i (t - \alpha_i)$ and $Q(t) = \sum_i \sigma_i D(t)/(t - \alpha_i)$. Direct substitution in (A3.2) gives

$$G(1, -t) = -F(t), \quad F = nDQ' - Q \sum_i (n - m_i) \frac{D}{t - \alpha_i}. \tag{A3.4}$$

Thus the form G , rather than the degree of a particular affine polynomial, records the whole critical divisor.

Theorem A3.1 (Finite flat reconstruction of the critical divisor). *Over $\mathcal{U}$, the zero scheme $\mathcal{Z} = V(G) \subset \mathbb{P}_{\mathcal{U}}^1$ is finite locally free of degree $2r - 3$, and is disjoint from $V(D_h Q_h)$. It is isomorphic, as a scheme over the base, to the entire reciprocal critical scheme on invertible pencil members. These assertions include nonreduced fibres and arbitrary base changes.*

Under a change of pencil frame the divisor transforms on the projective pencil line; under simultaneous congruence it is unchanged up to the induced isomorphism. For a family specified by its framed first relation spaces, with split semisimple spectral data and data matrices as above, the divisor is consequently reconstructed from its order- d failure neighbourhoods. Here $d = n^2 + 2n - 4$, and the supplied data and semisimple condition are additional to, not restrictions on, the all-pencil inverse theorem.

Proof. At a zero of L_i the residue of ω is $n - m_i$; at a zero of Q_h it is $-n$. These are nonzero units in characteristic zero. All the poles are simple and mutually disjoint on $\mathcal{U}$. Hence in every geometric fibre the numerator G is nonzero at every pole. This proves disjointness as schemes: a nonempty intersection would have a geometric point in some fibre. It also proves that G is not identically zero in any fibre.

Here is an elementary finite-flat argument which also deals with arbitrary, possibly nonreduced, bases. At each point of $\mathcal{U}$, choose a constant rational point of the projective line outside the finite zero set of its fibre of G . The characteristic-zero residue field contains its prime field $\mathbb{Q}$. Thus $\mathbb{P}^1(\mathbb{Q})$ supplies infinitely many constant points, whereas the nonzero fibre form has only finitely many zeros. One of these points avoids them. Its evaluation is nonzero in the residue field, and becomes a unit after shrinking the base. A constant projective coordinate change takes this point to infinity. The scheme $\mathcal{Z}$ is then contained in the complementary affine chart, where its equation has unit leading coefficient and can be made monic of degree $2r - 3$. Its algebra is free on $1, t, \dots, t^{2r-4}$. These neighbourhoods cover the base, proving finite local freeness and the stated degree. The same presentations survive all base changes. They also identify $\mathcal{Z}$ as a relative effective Cartier divisor; compare the general criterion in [2, Tag 062Y].

We next prove the identification with the critical scheme, rather than only a bijection of closed points. Locally trivialize the projective direction by a nonsingular representative $K_0(t)$ and write $K = \lambda K_0(t)$, where λ is invertible. Let $U_0(t) = \text{tr}(SK_0(t)^{-1})$. The rational differential of the reciprocal likelihood is obtained from

$$-n \log \lambda - \log \det K_0(t) - U_0(t)/\lambda.$$

Its radial score is $(-n\lambda + U_0)/\lambda^2$. Thus the critical coordinate ring imposes exactly $\lambda = U_0/n$, and makes U_0 a unit. Eliminating λ from the remaining score gives

$$-d \log \det K_0 - n d \log U_0 = \omega.$$

The determinant factorization and $U_0 = Q_h/D_h$ give the last equality. Since $\mathcal{Z}$ is disjoint from $D_h Q_h$, all divisions are by units on $\mathcal{Z}$. These are inverse coordinate-ring homomorphisms, including nilpotents. Globally the inverse sends a direction represented by K_0 to

$$K = \frac{U(K_0)}{n} K_0. \tag{A3.5}$$

Replacing K_0 by aK_0 replaces $U(K_0)$ by $a^{-1}U(K_0)$, so this formula glues. It also proves that no projective chart has been omitted.

Both $-d \log \det K_0 - n d \log U(K_0)$ and (A3.5) are invariant under pencil reparametrization. Simultaneous congruence preserves the trace and changes the determinant by a constant square; its relative logarithmic differential is therefore unchanged. Finally apply Theorem 10.3 and the universal construction in Theorem A2.1. This last reconstruction step uses a framed classifying family; it asserts neither an equivalence of unmarked moduli stacks nor recovery of a data matrix from an unmarked complex scheme alone. $\square$

A3.2. Hessian degeneracy as scheme-theoretic ramification. Write $\pi : \mathcal{Z} \rightarrow \mathcal{U}$. The relative ramification scheme is defined by $\text{Fitt}_0 \Omega_{\mathcal{Z}/\mathcal{U}}$. The trace discriminant on the base is the determinant of $(a, b) \mapsto \text{Tr}_\pi(ab)$ on $\pi_* \mathcal{O}_{\mathcal{Z}}$; it is an ideal locally generated by a section of an invertible sheaf, not a choice of eigenbasis.

Theorem A3.2 (Ramification, Hessians, and the trace discriminant). *Under Theorem A3.1, the ramification ideal is exactly the ideal generated by the determinant of the reciprocal score Hessian on the critical scheme. The trace-discriminant ideal is locally generated, up to a unit, by the binary discriminant $\text{Disc}(G)$. Both ideals commute with arbitrary base change. The finite critical scheme is etale precisely off this trace discriminant, and the totally real cover of Theorem A2.2 is a restriction of this finite flat family.*

Proof. On a base neighbourhood with the monic presentation $\mathcal{A} = \mathcal{O}_{\mathcal{U}}[t]/(g(t))$ used above,

$$\Omega_{\mathcal{A}/\mathcal{O}_{\mathcal{U}}} = \mathcal{A} dt / (g'(t) dt).$$

Its zeroth Fitting ideal is (g') . In the local coordinates (λ, t) of the preceding proof the radial second derivative, after imposing the radial score, is $-n/\lambda^2$, a unit. The complementary Schur factor of the Hessian is the derivative of the eliminated score. That score is $v(t)g(t)$, with v a unit along the critical scheme, since its only denominators come from the disjoint poles. Its derivative modulo (g) is vg' . The Hessian determinant thus generates (g') . A change of smooth pencil coordinates multiplies this determinant on the critical scheme by the square of a Jacobian unit. This proves the claimed equality of ideal sheaves independently of the coordinates and without reducing the scheme.

In the basis $1, t, \dots, t^{m-1}$, $m = 2r - 3$, the determinant of the trace pairing for a monic polynomial algebra is $\text{disc}(g)$. One proof checks the Vandermonde identity for the polynomial with independent roots and then descends the resulting polynomial identity in its coefficients; consequently it remains valid for repeated roots and arbitrary coefficient rings. Making the leading coefficient a unit and changing projective frame multiply the binary discriminant by units. Hence these local ideals are exactly $(\text{Disc}(G))$. This uses the classical trace-discriminant construction, not a new general discriminant theorem; see [2, Tags 0BVH, 0BJF].

The differential presentation, its Fitting ideal, and the matrix of the trace pairing all commute with base change. A monic algebra is etale exactly when g' is a unit in it, equivalently when its discriminant is a unit. On the separated real chamber the sign argument in Theorem A2.2 proves this condition and labels all m real roots. Its additional affine leading-coefficient localization merely selects one chart of the present family. No new sign argument is needed. $\square$

The trace discriminant on the base is not being identified with an arbitrarily scheme-theoretically imaged ramification locus: their multiplicities need not coincide. Likewise finite flatness is asserted on $\mathcal{U}$, not over data for which a zero of Q_h collides with a pole or becomes multiple. These exclusions concern the spectral data polynomial, not collisions of zeros of G .

Coordinate invariance on the critical quotient. Write a smooth local coordinate change as $x = x(y)$, its Jacobian as $J = (\partial x_a / \partial y_i)$, and the first scores as s_a . The second-derivative chain rule is

$$H_y = J^t H_x J + \sum_a s_a \left(\frac{\partial^2 x_a}{\partial y_i \partial y_j} \right)_{ij}.$$

All s_a vanish in the critical quotient itself, not only on its reduction. There $H_y = J^t H_x J$ and $\det H_y = (\det J)^2 \det H_x$. Since $\det J$ is a unit, the Hessian ideal is unchanged even in a nonreduced critical algebra.

A3.3. A length-three critical fibre at a nonsingular pencil member.

Proposition A3.3 (A triple critical collision in a fixed pencil). *Let $A = I_3$, $B = \text{diag}(-1, 0, 1)$, and*

$$S_c = \text{diag}((1+c)/2, -c, (1+c)/2), \quad c \in \mathbb{C} \setminus \{0, -1\}.$$

The reciprocal critical schemes form a finite flat family of length three. At $c = 1$ all three critical points are real and reduced; one lies in the projective direction $x = 0$. At $c = -2/3$ the full critical scheme is supported at $K = I_3/3$ and is isomorphic to $\text{Spec } \mathbb{C}[u]/(u^3)$. The trace-discriminant ideal of this family is generated, up to a unit, by $(3c+2)^3$.

Proof. Here $D(t) = t(t^2 - 1)$, $Q(t) = t^2 + c$, and

$$F(t) = -(6c+4)t^2 + 2c. \tag{A3.6}$$

The excluded parameters are exactly those where Q is repeated or meets a zero of D . Thus this entire family lies in $\mathcal{U}$. Homogenizing F to its projective degree three gives $\widehat{F}(T, Z) = -(6c+4)T^2Z + 2cZ^3$. The associated projective direction is $[x : y] = [Z : -T]$. At $c = 1$ its zeros are $t = \pm 1/\sqrt{5}$ and the point $[T : Z] = [1 : 0]$, all simple. Formula (A3.5) takes the latter to $I_3/3$.

At $c = -2/3$, $\widehat{F} = -4Z^3/3$, so its whole zero scheme is in the chart $T = 1$, where $u = Z/T = -x/y$. The local equation is $-4u^3/3$. The radial reconstruction is an isomorphism on this nonreduced scheme and its reduced point is $I_3/3$, since $\text{tr } S_c = 1$. The binary cubic discriminant, or the trace calculation in a monic chart, is

$$\text{Disc}(\widehat{F}) = 64c(3c+2)^3. \tag{A3.7}$$

Since c is a unit on the stated base, this gives the claimed ideal. The special ramification ideal is (u^2) in $\mathbb{C}[u]/(u^3)$, in agreement with the Hessian theorem. $\square$

This example keeps the pencil and its spectral sheaf fixed. It shows why the critical correspondence must be retained with its scheme structure: the number of distinct critical points drops while the finite flat degree stays three. The order- d inverse theorem, followed by the supplied data family, reconstructs this collision and its trace-discriminant multiplicity. This is a structural consequence of the same reconstruction chain, not a replacement for the independent real-root argument or a claim that generic degree formulas are new.

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REFERENCES

- [1] C. Fevola, Y. Mandelshtam, and B. Sturmfels, Pencils of quadrics: old and new, *Le Matematiche* **76** (2021), no. 2, 319–335. doi:10.4418/2021.76.2.2; arXiv:2009.04334v2, Theorem 1.1, Corollary 2.1, Remark 2.3, Conjecture 4.5, and Theorem 5.1.
- [2] The Stacks Project Authors, *The Stacks Project*, Tags 056Q, 062Y, 0BVH, and 0BJF, relative effective Cartier divisors and discriminants of finite locally free morphisms; accessed September 24, 2026. <https://stacks.math.columbia.edu>.